Predicting Antibody-Antigen Binding from Sequence: A Comparison of Deep Learning Classifiers on Protein Language Model Embeddings

Candidate Numbers: 63600, 74166, 59515

Abstract

This project aims to predict whether an antibody sequence will bind to a given antigen target, a central problem in computational antibody design and therapeutic development. Current approaches to binding prediction either rely on expensive physics-based simulations or require solved three-dimensional structures that are unavailable for most candidate sequences. It remains unclear which neural architecture provides the most suitable inductive bias for modelling antibody-antigen interactions from frozen protein language model embeddings. We propose a sequence-only approach that extracts learned representations from a pretrained protein language model (ESM-2) and feeds them into supervised classifiers to predict binding. The project contributes three innovations: (1) a systematic comparison of MLP, CNN, LSTM, GRU and Transformer classifiers on ESM-2 embeddings for binding prediction; (2) a novel cross-attention architecture that explicitly models directional antibody-antigen interaction, achieving a mean AUROC of 0.795 ± 0.023; and (3) an evaluation of how predictive performance varies across model architectures of increasing complexity.

1 Introduction

Antibodies are proteins produced by the human body which bind to specific antigens, where antigens are structures found on the surface of pathogens (such as bacteria and viruses), triggering an immune response. Due to the role of antibodies in immune defence, they have become a crucial tool in the treatment of a range of diseases. The effectiveness of an antibody is predominantly dependent on its binding affinity to the target antigen so obtaining a measurement of this is essential in antibody development.

The traditional approach of experimentally measuring the affinity of candidate antibodies to their targets is time-consuming and resource-intensive. Incorporating computational methods improves overall efficiency. However, the current strategies encounter some problems. Physics-based simulations operate by modelling the interactions of molecules. Although this typically produces accurate results, it is computationally demanding, making it impractical when going through large-scale testing of candidate sequences of antibodies. Alternatively, there are structure-based methods that require pre-solved three-dimensional representations of the molecules. Given the large number of possible combinations, only a small fraction of the sequences have the solved structures readily available for processing.

Therefore, there is increased interest in the use of deep learning methods, specifically those which are sequence-based, as they directly use the sequences rather than relying on structural data. Our project and paper investigates such an approach using ESM-2, a transformer-based protein language model, pretrained on 250 million protein sequences using a masked language modelling objective, analogous to BERT in natural language processing. We evaluate the use of MLP, CNN, LSTM, GRU and cross-attention Transformer classifiers on the resulting embeddings, and demonstrate that cross-attention over protein language model embeddings outperforms all other architectures tested.

In summary, this paper makes the following contributions:

1. A systematic comparison of MLP, CNN, LSTM, GRU and Transformer classifiers on ESM-2 embeddings for binding prediction.

2. A novel cross-attention Transformer architecture that explicitly models directional antibody-antigen interaction, achieving a mean AUROC of 0.795 ± 0.023.

3. An empirical finding that cross-attention suggests a stronger inductive bias in this setting than all other architectures tested for protein binding prediction.

To our knowledge, this is the first systematic comparison of MLP, CNN, LSTM, GRU, and cross-attention Transformer classifiers on frozen ESM-2 embeddings for antibody-antigen binding prediction.

2 Related Work

2.1 Protein Language Models

Sequence-based prediction tasks have benefitted from the development of large-scale protein language models. Rives et al. (2021) demonstrated that transformer-based models trained on hundreds of millions of protein sequences using self-supervised objectives learn rich representations encoding evolutionary, structural, and functional information without requiring labelled data. Building on this, Lin et al. (2023) introduced ESM-2, scaling this approach to 650 million parameters and demonstrating state-of-the-art performance across a range of downstream protein prediction tasks including structure prediction, function annotation, and mutational effect prediction. The representations learned by ESM-2 capture residue-level structural and functional information directly from sequence, making them well-suited as fixed feature extractors for downstream classification tasks. Our work adopts this by using frozen ESM-2 650M embeddings as input to supervised classifiers.

2.2 Attention Mechanisms and Transformers

Vaswani et al. (2017) introduced the Transformer architecture, replacing recurrent connections with self-attention mechanisms that directly model pairwise relationships between all positions in a sequence, regardless of distance. This is particularly relevant for protein sequences where residues far apart in the sequence may be spatially adjacent in the folded structure and critical for binding function. Devlin et al. (2019) extended this framework with BERT, demonstrating that bidirectional transformer models pretrained using masked token prediction learn transferable representations applicable to diverse downstream tasks, which inspired the ESM-2's pretraining objective. Bahdanau et al. (2015) introduced attention-based alignment in neural machine translation, demonstrating that attention weights provide interpretable evidence for which input positions drive model predictions. In our cross-attention Transformer, the antibody embedding attends directly to the antigen embedding and vice versa, directly operationalising this principle for binding prediction.

2.3 Deep Learning for Binding Prediction

He et al. (2016) demonstrated that residual connections provide a powerful architectural inductive bias, motivating our inclusion of residual connections in the cross-attention Transformer. Sequential models such as LSTM and GRU have been widely applied to biological sequence tasks, exploiting the natural ordering of residues. Convolutional approaches have been applied to detect local motifs in sequences, while Transformer-based methods have increasingly demonstrated advantages in long-range interaction tasks. To our knowledge, no prior work has systematically evaluated MLP, CNN, LSTM, GRU, and cross-attention Transformer classifiers on frozen ESM-2 embeddings for antibody-antigen binding prediction on SAbDab, making direct numerical comparison with existing literature difficult. Our cross-attention architecture achieves a mean AUROC of 0.795, providing a strong baseline for future work in this direction. Existing sequence-based approaches include AbAgIntPre (Huang et al., 2022), which uses a Siamese CNN with k-spaced amino acid pair encoding, and AttABseq (Jin et al., 2024), an attention-based model for affinity prediction from sequence alone. Unlike these approaches, our work leverages pretrained protein language model embeddings from ESM-2 as a fixed feature extractor, avoiding hand-crafted sequence encodings entirely.

3 Data Description

3.1 Data Sources

We construct a dataset of antibody-antigen binding pairs using the Structural Antibody Database (SAbDab) (Dunbar et al., 2014), which is a curated repository of experimentally resolved antibody-antigen complexes from the Protein Data Bank (PDB). After filtering for protein-protein complexes that have complete heavy and antigen chain annotations, we obtain 2,692 unique complexes that span diverse antibody germline families and antigen targets.

Table 1: Dataset split statistics

Split	Total Pairs	Positive	Negative
Train	4,307	2,154	2,154
Validation	538	269	269
Test	539	269	270
Total	**5,384**	**2,692**	**2,692**

Positive pairs were extracted directly from SAbDab experimentally verified complexes. Negative pairs were generated by randomly mismatching antibodies with non-cognate antigens, following standard practice in literature. Stratified splitting ensures class balance is preserved across all splits, preventing optimistic bias in performance estimates.

3.2 Sequence Representation

Each antibody heavy chain and antigen sequence is represented as a dense embedding vector with dimension 1,280, extracted from the final layer of ESM-2 650M (Lin et al., 2023), which is a transformer-based protein language model that is pretrained on 250 million protein sequences using a masked language modelling objective. Mean pooling is applied over the sequence length dimension, producing a fixed length representation per protein. Sequences exceeding the ESM-2 maximum context are truncated to a length of 1,022.

4 Methodology

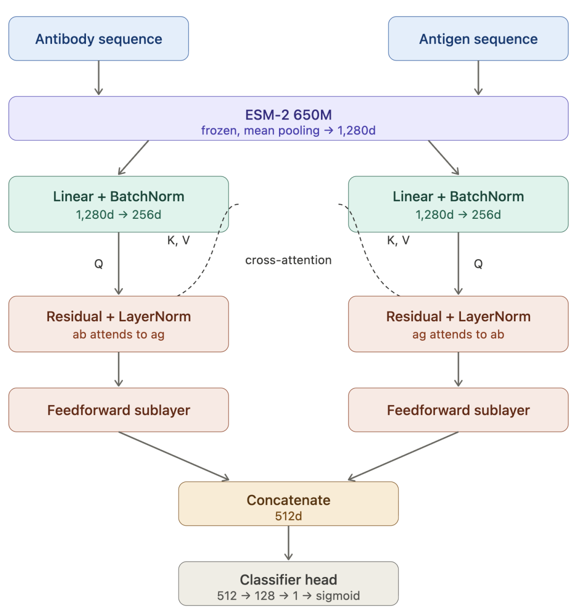

Figure 1: Architecture of the cross-attention Transformer classifier. Antibody and antigen sequences are embedded using frozen ESM-2 650M, projected to dimension 256, then processed through bidirectional cross-attention before concatenation and binary classification.

All models are implemented in PyTorch and can be trained on a GPU preferably (T4 was used) or a CPU. Models are trained on 5,384 antibody-antigen pairs split into training (80%), validation (10%), and test (10%) sets using stratified sampling to maintain class balance. Input representations are extracted from the final layer of ESM-2 650M and mean-pooled over the sequence dimension to produce fixed-length embeddings of dimension 1,280 per protein. Each architecture embodies a different inductive bias: the MLP makes no structural assumptions about the relationship between embeddings; the CNN assumes local spatial interactions in the feature interaction matrix; the LSTM and GRU assume sequential dependencies between the antibody and antigen tokens; and the cross-attention Transformer assumes global pairwise interactions between the two protein representations. Our experimental design tests which of these inductive biases best captures antibody-antigen binding.

This comparison functions as a controlled ablation over interaction mechanisms: the MLP serves as a concatenation-only baseline with no interaction structure; the CNN introduces local pairwise interaction modelling; the LSTM and GRU impose sequential structure over the token pair; and the cross-attention Transformer enables explicit bidirectional alignment between protein representations. By fixing all other training conditions, performance differences can be attributed to architectural inductive bias alone.

The MLP classifier serves as our non-sequential baseline. Our mean pooled ESM-2 embeddings of the antibody-antigen sequences are concatenated and formed into a fixed-length input vector with a dimension of 2,560.

$$\mathbf{x} = [\mathbf{e}_{ab} \| \mathbf{e}_{ag}] \in \mathbb{R}^{2560} \tag{1}$$

It is then passed through a fully connected hidden layer of dimension 512 with ReLU activation, batch normalisation and dropout regularisation, followed by a second linear projection to dimension 256 with ReLU activation and dropout, before a sigmoid output to produce a scalar binding probability. The model is trained using an Adam optimiser with a learning rate of 1 x 10^-3. Adam is used as the optimiser as its adaptive learning rates are well-suited to small datasets where gradients may be noisy. L2 weight decay 1 x 10^-4 with Binary Cross Entropy loss for binary classification is also used. Binary Cross Entropy is appropriate here as binding prediction is a binary outcome: an antibody either binds or does not.

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{2}$$

Early stopping is applied with a patience count of 10 epochs, and terminates if validation AUROC is not better than the previous validation AUROC for 10 consecutive epochs.

To model the relationship between antibody and antigen embeddings, we compute an outer product interaction matrix via batched matrix multiplication, producing a 128×128 matrix where position (i,j) represents the interaction between antibody feature i and antigen feature j. This interaction matrix is then treated as a single-channel image and processed by a 2D CNN. Two 2D convolutional layers with 32 and 64 filters respectively, kernel size 3, batch normalisation, ReLU activation, max pooling, and dropout regularisation (p=0.3) are applied over the interaction matrix. Global average pooling collapses the spatial dimensions before a fully connected classifier head with hidden dimension 64, ReLU activation, dropout, and sigmoid output produces a binding probability. The same Binary Cross Entropy loss, Adam optimiser, and early stopping configuration as the MLP are used.

The LSTM classifier processes the antibody and antigen embeddings as a two-token sequence, following the stacked LSTM architecture. The antibody embedding is presented as the first token and the antigen as the second, forming a sequence of length two. This is passed through a two-layer bidirectional LSTM with a hidden dimension of 128, where lower layers learn local feature representations and upper layers learn higher-level patterns. Inter-layer dropout regularisation (p=0.3) is applied between LSTM layers. The concatenated forward and backward hidden states from the final time step, of dimension 256, are passed through a fully connected classifier head with a hidden dimension of 128, ReLU activation, dropout regularisation, and a sigmoid output unit. The same Binary Cross Entropy loss, Adam optimiser, and early stopping configuration as the MLP are used. A bidirectional architecture is used so that the model processes the sequence in both directions (antibody to antigen and antigen to antibody) producing a more symmetric representation of the interaction.

The GRU classifier uses an identical architecture to the LSTM with nn.GRU substituted for nn.LSTM. The GRU simplifies the LSTM gating mechanism by merging the forget and input gates into a single update gate, reducing the parameter count while maintaining comparable representational capacity. This makes it a more parameter-efficient alternative, particularly relevant for our relatively small dataset. All other architectural choices, such as hidden dimension, number of layers, dropout, classifier head, and training configuration, are kept identical to the LSTM to enable a clean architectural comparison.

$$\text{CrossAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V \tag{3}$$

The cross-attention Transformer explicitly models the directional interaction between antibody and antigen, following the cross-attention mechanism of Vaswani et al. (2017) and the BERT classification approach of Devlin et al. (2019). Both embeddings are first projected to a dimension of 256 from separate linear layers with batch normalisation. The antibody embedding then serves as the query attending to the antigen as key and value, and vice versa, directly modelling which features of each protein are most relevant to the other, as defined in Equation 3. Residual connections and layer normalisation are applied after each cross-attention operation, followed by feedforward sublayers. The resulting antibody and antigen representations are concatenated and passed through a classifier head with hidden dimension 128, ReLU activation, dropout regularisation (p=0.1), and a sigmoid output.

Algorithm 1 Cross-Attention Transformer Training

Require: Training pairs $\{(\mathbf{e}_{ab}^{(i)}, \mathbf{e}_{ag}^{(i)}, y^{(i)})\}_{i=1}^{N}$
Ensure: Trained parameters θ
Initialise parameters θ randomly
for each epoch **do**
  for each mini-batch $(\mathbf{e}_{ab}, \mathbf{e}_{ag}, \mathbf{y})$ **do**
    Project: $\mathbf{ab} = \text{BatchNorm}(W_{ab} \cdot \mathbf{e}_{ab})$
    Project: $\mathbf{ag} = \text{BatchNorm}(W_{ag} \cdot \mathbf{e}_{ag})$
    Cross-attend: $\mathbf{ab}' = \text{CrossAttn}(Q{=}\mathbf{ab},\ K{=}\mathbf{ag},\ V{=}\mathbf{ag})$
    Cross-attend: $\mathbf{ag}' = \text{CrossAttn}(Q{=}\mathbf{ag},\ K{=}\mathbf{ab},\ V{=}\mathbf{ab})$
    Residual: $\mathbf{ab}_{out} = \text{LayerNorm}(\mathbf{ab} + \mathbf{ab}')$
    Residual: $\mathbf{ag}_{out} = \text{LayerNorm}(\mathbf{ag} + \mathbf{ag}')$
    Classify: $\hat{y} = \sigma(\text{MLP}([\mathbf{ab}_{out} \| \mathbf{ag}_{out}]))$
    Compute loss: $\mathcal{L} = \text{BCE}(\hat{y}, \mathbf{y})$
    Update $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$
  end for
  if validation AUROC does not improve for 10 epochs **then**
    break ▷ Early stopping
  end if
end for

5 Experiments

We evaluate all models on the test set of 539 pairs. Primary metrics are AUROC, AUPRC, and F1 score at a 0.5 classification threshold. AUROC measures overall discrimination ability, AUPRC is more informative under class balance, and F1 captures the harmonic mean of precision and recall. All models use identical training configurations for the Adam optimiser, a learning rate of 1×10^{-3}, L2 weight decay of 1×10^{-4}, Binary Cross Entropy loss, and early stopping with patience 10.

Hyperparameter	Value
Optimiser	Adam
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Loss function	Binary Cross Entropy
Batch size	32
Max epochs	50
Early stopping patience	10
Dropout	0.3 (MLP, CNN, LSTM, GRU) / 0.1 (Transformer)
Embedding dimension	1,280

Table 2: Shared hyperparameters across all models.

Hyperparameters were selected based on standard values from related work, with learning rate and dropout verified on the validation set. All models were evaluated across three independent runs with seeds 42, 123 and 456, with mean and standard deviation reported.

6 Numerical Evaluation

Table 3: Statistical comparison of models

Model	AUROC	AUPRC	F1
Reference			
Random classifier	0.500	0.500	—
Baselines			
MLP	0.694 ± 0.008	0.684 ± 0.004	0.686 ± 0.010
CNN	0.737 ± 0.022	0.709 ± 0.022	0.680 ± 0.026
Sequentials			
LSTM	0.618 ± 0.117	0.600 ± 0.114	0.694 ± 0.039
GRU	0.781 ± 0.017	0.767 ± 0.011	0.736 ± 0.030
Attention-Based			
Cross-Attention Transformer	0.795 ± 0.023	0.774 ± 0.010	0.736 ± 0.018

GRU outperforms LSTM with mean AUROCs of 0.781 and 0.618 respectively. The LSTM exhibits high variance across runs (± 0.117), suggesting sensitivity to initialisation on a dataset of this size. GRU's simpler update gate reduces this risk while maintaining comparable representational capacity. The MLP and CNN achieve mean AUROCs of 0.694 and 0.737 respectively, suggesting that the outer product interaction matrix provides limited additional signal over direct concatenation on this dataset. The cross-attention Transformer outperforms all other architectures with a mean AUROC of 0.795 ± 0.023, leading the next best model (GRU, 0.781) by 0.014 AUROC points, suggesting that explicitly modelling the directional interaction between antibody and antigen is the key inductive bias for binding prediction from protein language model embeddings. All models learn meaningfully above the random baseline of 0.5, confirming that frozen ESM-2 embeddings contain sufficient information for above-chance binding prediction. Notably, GRU outperforms CNN (0.781 vs 0.737) despite operating on a two-token sequence rather than an interaction matrix, suggesting that sequential processing of the embedding pair captures complementary information to spatial interaction modelling.

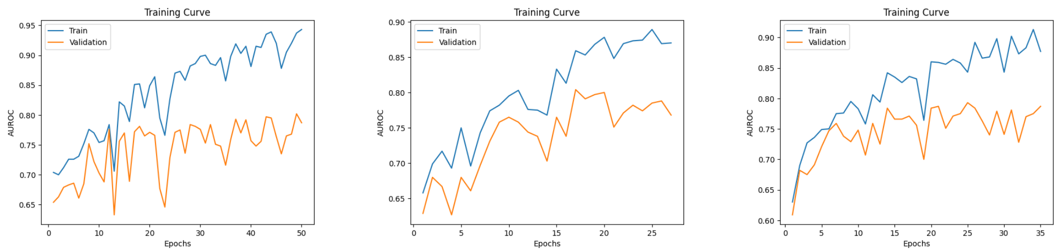

Figure 2: Training curve of the Cross-Attention Transformer across 50 epochs using 3 random seeds.

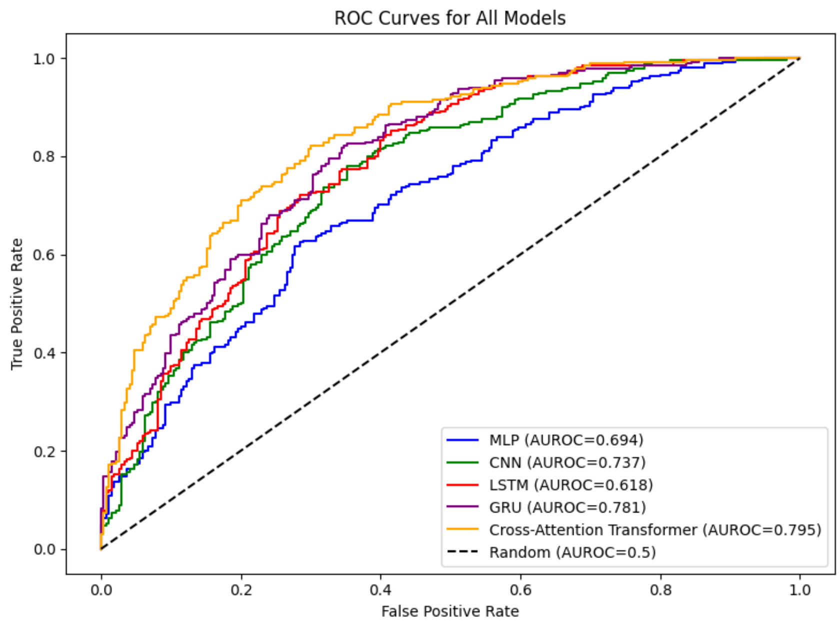

Figure 3: ROC curves for all five classifiers evaluated on the test set. Cross-attention Transformer achieves the highest mean AUROC of 0.795 ± 0.023, demonstrating the advantage of explicit interaction modelling over all other architectures.

7 Conclusion

In this paper we presented a systematic comparison of five deep learning architectures for antibody-antigen binding prediction using frozen ESM-2 650M protein language model embeddings. Our results demonstrate a clear hierarchy of performance: cross-attention outperforms all other approaches, with GRU as the next best model.

The MLP and CNN baselines achieve AUROCs of 0.694 ± 0.008 and 0.737 ± 0.022 respectively, establishing that ESM-2 embeddings alone contain sufficient information for above-chance binding prediction. The GRU outperforms CNN with a mean AUROC of 0.781, while the LSTM showed high variance across runs (mean AUROC 0.618 ± 0.117), indicating sensitivity to random initialisation on small datasets. The cross-attention Transformer achieves an AUROC of 0.795 ± 0.023, outperforming all other architectures. This result suggests that explicitly modelling the directional interaction between antibody and antigen, is the key inductive bias for binding prediction from protein language model embeddings.

Our study has several limitations. The dataset of 2,692 positive pairs from SAbDab is relatively small, which may limit the diversity of antibody-antigen interactions represented. Mean pooling collapses each sequence to a single vector before the classifier, discarding per-residue information and making attention weight analysis uninformative in the current setup; the cross-attention operates over single-token representations with no positional distinctions to recover. This likely constrains the benefit of attention-based architectures, as residue-level signal is removed prior to modelling interaction. Negative pairs were generated by random mismatching rather than experimentally verified non-binders, which may not fully reflect the true distribution of non-binding pairs. Finally, compute constraints on our training environment limited the scope of hyperparameter search. Additionally, the antibody representation uses only the heavy chain sequence, discarding the light chain which contributes to the binding interface in many complexes.

Future work could explore per-residue ESM-2 embeddings to enable residue-level cross-attention and CDR analysis, where attention weights would then provide interpretable evidence for which residues drive binding, building on the alignment principle of Bahdanau et al. (2015). Additional directions include evaluation on the full SAbDab database, fine-tuning of ESM-2 weights alongside the classifier head, and extension to affinity regression using binding measurements available in SAbDab for a subset of complexes.

These results suggest that sequence-only cross-attention classifiers could serve as a practical first-pass screening tool in antibody drug development pipelines, reducing reliance on expensive structural methods for initial candidate filtering.

8 References

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9 Statement about Individual Contributions

All three group members contributed equally to this project. Each member contributed 33% to the overall work, encompassing data preprocessing, model implementation, evaluation, and paper/slide writing.

Candidate Number	Contribution
63600	33%
74166	33%
59515	33%